

Review v16 and genuine priors for the normal-crossing example

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Abstract

Review v16 (independent, statement level) found Headlines XX, XX', XX'' of units 218–220 *faithful with caveats*: the kernel identification, the constant $2\rho(0)\sqrt{2\pi}e^{z^2/2}$ (checked independently through the product coordinate $t = x_0x_1$, whose pushforward density is $2\log(1/|t|)$), the tilt and the three modes of convergence are right; the should-fix items are (i) the Lean weights may be signed, so “posterior” needs $\rho \geq 0$; (ii) the statements are fixed- θ moment generating function limits, not weak convergence of posterior measures; (iii) no $O(1/\log n)$ rate is proved; (iv) “unconditional” must distinguish XX (assumes $Z_n \rightarrow z$) from XX'/XX''; (v) “variance 1 for every $n \geq 1$ ”; (vi) the tightness argument uses zero means, unit variances and pairwise independence, not “second moments only”. Unit 221 supplies the one Lean item: for a genuine prior weight ($\rho \geq 0$ on the box, $\rho(0) > 0$) the evidence $\int_{[-1,1]^2} \rho L_n$ is strictly positive at every sample size (`nc_evidence_pos`) and the posterior mean of 1 is 1 (`ncPosteriorMean_one`); the wording items were applied to the module headers, the mirror, the reports and the headline index. Zero sorry/axiom; 9098 jobs.

1 The things formalised

```
theorem nc_integral_pos (h : Continuous ) (h nn : x symBox 2, 0 x) (h 0 : 0 < 0)
(N z) : 0 < x in symBox 2, x * ncKernel N z x
theorem nc_evidence_pos (hn : 0 < n) ... : 0 < x in symBox 2, x * ncLikelihood n y x
theorem ncPosteriorMean_one (hn : 0 < n) ... : ncPosteriorMean n y (fun _ => 1) = 1
```

2 Mathematics

Continuity gives $\rho \geq \rho(0)/2$ on a sup-norm ball B_δ around the origin, $\delta \leq 1$, which lies in the box; there the kernel is at least $e^{-N^2/2-|Nz|}$ since $|x_0x_1| \leq 1$. So $\int_{[-1,1]^2} \rho K \geq \int_{B_\delta} \rho K \geq \frac{\rho(0)}{2} e^{-N^2/2-|Nz|} |B_\delta| > 0$, using nonnegativity of the integrand on the whole box for the monotonicity in the domain. The truth’s likelihood factor is a positive constant.

3 Lean notes

- `setIntegral_ge_of_const_le` now returns `.real s • c` ; convert with `smul_eq_mul`, `measureReal_def`.
- `rw [abs_mul]` rewrites the first matching product; pass the factors explicitly (`abs_mul (N * z) (x 0 * x 1)`) when two products are present.
- `setIntegral_mono_set` wants the set inclusion as `s [] t`: `Filter.Eventually.of_forall hball`.

- Reflow docstring paragraphs line-count-neutrally when downstream **#Lnn** dots must stay valid.

4 Status

H2-lite complete and reviewed (Headlines XX–XX'' plus the genuine-prior lemmas). Per Astra #24: freeze; E2 (Abelian Mellin coefficient) remains an optional appendix.